ENVIRONMENTAL STUDIES REPORT

MANOHAR PARRIKAR INTEGRATED SOLID WASTE MANAGEMENT (ISWM) FACILITY, CACORA, GOA

1. Executive Profile & Plant Overview:-

The **Manohar Parrikar Integrated Solid Waste Management (ISWM) Facility** at Cacora, South Goa, is a flagship regional waste management project established by the **Goa Waste Management Corporation (GWMC)** under the Department of Science, Technology and Waste Management, Government of Goa. Constructed by **SMC Infrastructure Pvt. Ltd.** and operated via **M/s Vasudha Waste Treatment Pvt. Ltd.**, the plant was built on an 81,000 m^2 former open dumping ground that was completely cleared using bioremediation before facility construction. Inaugurated by Prime Minister Narendra Modi on February 6, 2024, the facility operates in full compliance with the **Solid Waste Management Rules, 2016** as a zero-liquid-discharge, resource-recovery complex.

Key Technical Specifications:-

- **Location:** Cacora, Kudchade (Curchorem), Quepem Taluka, South Goa, Goa.
- **Operational Capacity:** 100 Tonnes Per Day (TPD) (+25% expandable margin).
- **Processing Division:** 60 TPD Wet/Organic Waste + 40 TPD Dry/Inorganic Waste.
- **Core Technologies:** High-efficiency Thermophilic Biomethanation (Organic Extruder system), Aerobic Composting, Refuse-Derived Fuel (RDF) Processing Line, and a 450 KLD Effluent Treatment Plant (ETP).
- **Capital Investment:** ₹173.98 Crores (funded via state contribution and NABARD RIDF loan).
- **Power Output:** 800 kW Biogas Engine (~17,000 units/day) + Solar Rooftop Installation (~1,600 units/day).

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2. Waste Generation & Regional Dynamics:-

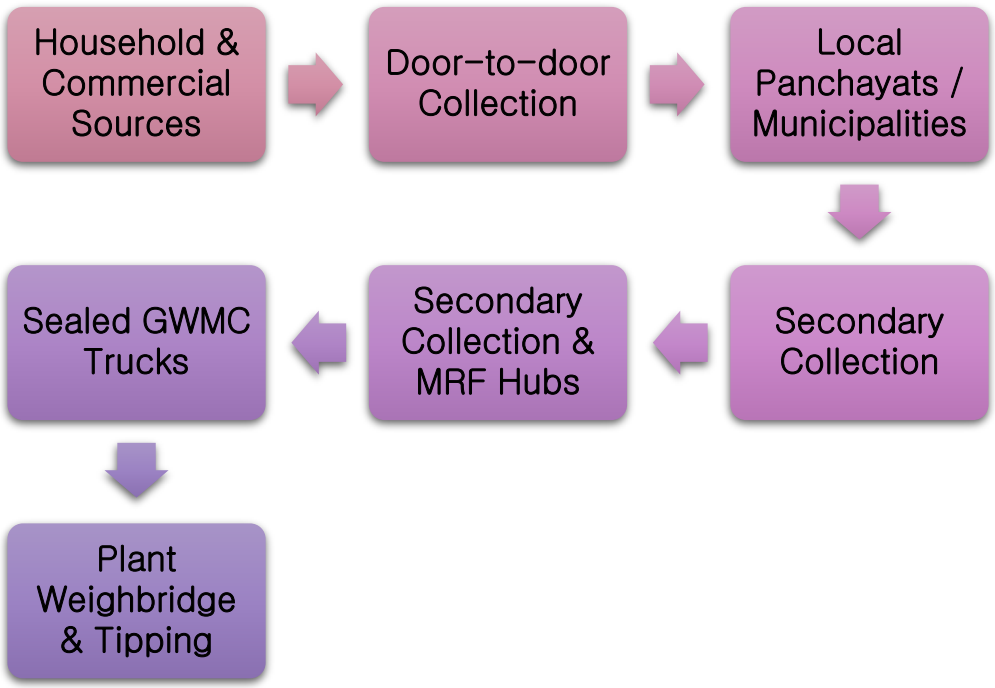
South Goa generates hundreds of tonnes of municipal solid waste daily across tourist hubs, residential areas, and rural panchayats. The Cacora facility caters to **4 Municipal Councils** (Curchorem-Cacora, Quepem, Sanguem, Canacona) and **30 Village Panchayats** across the talukas of Quepem, Sanguem, Dharbandora, and Canacona.

SOUTH GOA MSW STREAM	COMPOSITION & CHARACTERISTICS	DIVERSION / TARGET OUTPUT
Wet / Organic Waste (~60%)	Kitchen waste, hotel food scraps, market organics, agricultural litter.	Biomethanation (Biogas/Power) & Organic Compost.
Dry Recyclable Waste (~30%)	PET bottles, rigid plastics, paper, metals, tetra packs, glass.	Recovered recyclables (22% yield) & Refuse-Derived Fuel (RDF).
Inert & Residue Material (~10%)	Street sweepings, fine silt, unprocessable inert debris.	Engineered Sanitary Landfill (Only 4-5% goes to landfill).

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3. Storage, Collection, & Regional Logistics:-

Logistics Network & Supply Flow-



 At-Source Segregation: Waste is segregated at the household and commercial level into wet (organic) and dry (inorganic) streams.

 Primary Collection: Local Municipal Councils (e.g., Curchorem-Cacora, Quepem, Sanguem, Canacona) and Village Panchayats perform door-to-door collection.

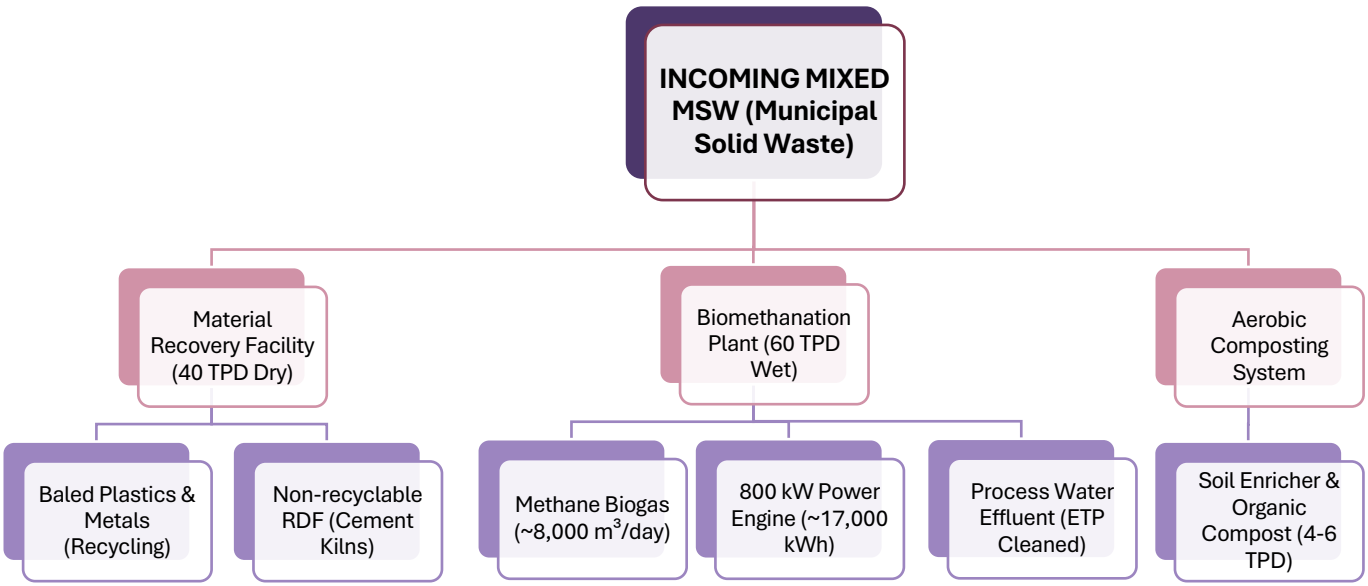
 Secondary Storage & GWMC Transport: Municipalities and Panchayats store segregated waste at designated secondary collection facilities. The GWMC deploys its fleet of specialized covered vehicles for regional long-distance transit.

 Plant Reception: Sealed trucks enter the Cacora plant, pass through an automated weighbridge, and unload waste into segregated processing bays.

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4. Processing Mechanics & Scientific Treatment:-

The plant integrates three primary technological pathways to recover resources and energy-



A. Material Recovery & Refuse-Derived Fuel (RDF) Facility-

Dry inorganic waste (40 TPD) undergoes mechanical sorting using trommels, ballistic separators, and magnetic separators alongside manual sorting bays. Recyclables (PET, HDPE, metals) are baled and dispatched to authorized recyclers. The non-recyclable high-calorific fraction is shredded to produce **Refuse-Derived Fuel (RDF)** for cement kilns and power generation.

B. High-Efficiency Biomethanation Unit (Wet Waste Processing)-

Segregated organic wet waste (60 TPD) is treated through thermophilic anaerobic digestion:

- **Biogas Generation:** Produces approximately 130–140 m^3 of biogas per tonne of wet waste (~8,000 m^3 total daily biogas yield with ~60% methane content).
- **Power Generation:** Biogas powers an 800 kW engine producing ~17,000 units of electricity daily. 40% is consumed in-house for plant operations, while 60% is exported to the state electricity grid.
- **Effluent Treatment:** Includes a 450 KLD Effluent Treatment Plant (ETP) to recycle and re-use process water, maintaining zero liquid discharge standards.

C. Aerobic Composting & Soil Conditioner Unit-

Stabilized organic digestate and residual sludge from the biomethanation process are cured, screened, and enriched to manufacture 4 to 6 tonnes per day of high-grade organic compost and soil conditioner for local agriculture and public parks.

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5. SDG Alignment & Impact Assessment:-

United Nations SDG	Facility Implementation & Environmental Contribution
SDG 3: Good Health and Well-being	Replaced unscientific open dumping at Cacora, preventing vector-borne diseases, soil toxicity, and hazardous fires.
SDG 7: Affordable and Clean Energy	Generates ~17,000 units of bio-electricity and ~1,600 units of solar energy daily, exporting surplus green energy to the grid.
SDG 9: Industry, Innovation, and Infrastructure	Incorporates advanced thermophilic digestion, automated sorting, and continuous oversight by experts from IIT Bombay, NEERI, and BARC.
SDG 11: Sustainable Cities and Communities	Solves waste handling challenges for over 30 village panchayats and 4 municipal councils across South Goa.
SDG 12: Responsible Consumption & Production	Achieves max circularity by diverting 90%+ of waste away from landfills into compost, recyclables, and RDF.
SDG 13: Climate Action	Captures methane (CH_4) gas from decomposing organic waste, significantly reducing direct greenhouse gas emissions.

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6. Visual Assets & Layout Schematic Descriptions:-



Figure 6.1: Aerial Overview of the Integrated Solid Waste Management Facility at Cacora, Goa



Figure 6.2: Material Recovery Facility (MRF) – Manual Sorting Conveyor Belt Operations



Figure 6.3: Thermophilic Anaerobic Digesters & Biogas Storage Domes

7. Conclusion:-

The **Manohar Parrikar Integrated Solid Waste Management Facility at Cacora** demonstrates how small towns and rural panchayats can implement sustainable waste handling. By reclaiming an 81,000 m^2 polluted dumpsite and replacing it with biomethanation, solar energy generation, zero liquid discharge ETP, and educational programs, Cacora serves as a model for decentralized, eco-friendly infrastructure in India.

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8. Official References & Source Links:-

- **Goa Waste Management Corporation (GWMC) — Official Cacora Facility Profile:**
https://gwmc.goa.gov.in/swmf-cacora/
Official Government portal with technical specifications, capital outlay, and concessionaire details.

- **Down To Earth / CSE Research Briefing — Cacora Milestone:**
https://www.downtoearth.org.in/waste/from-garbage-to-electricity-goa-reaches-new-waste-management-milestone
Verified analytical report detailing the 1,400 students trained (SDG 4), 450 KLD ETP (SDG 6), 4-5% landfill rate (SDG 12), and power output.

- **Goa Waste Management Corporation (GWMC) — Facilities Portal:**
https://gwmc.goa.gov.in/solidwaste/
Official state repository for integrated solid waste processing facilities across Goa.

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